

Employee Wellness Management Analytics

Milestone 3 – Emotion Detection & Journal Analytics

Project Objective

Develop an AI-powered Employee Wellness Management system that analyzes employee journal entries using multilingual NLP, transformer-based emotion detection, and VADER sentiment analysis. The system predicts emotions, calculates sentiment scores, provides wellness recommendations, and securely stores journal data.

Model Used

Hugging Face Transformer Model, VADER Sentiment Analyzer, PyTorch, spaCy, Google Translator, LangDetect.

Emotion Detection Pipeline

Journal Entry → Language Detection → NLP Preprocessing → Translation → Emotion Detection → Confidence Score → Sentiment Analysis → Wellness Recommendation → Database Storage → Frontend Display.

Confidence Score

The confidence score is the probability assigned by the transformer model to the predicted emotion.

Sentiment Analysis

Calculates Positive, Negative, Neutral, and Compound scores using VADER.

Database Schema

User ID, Journal Text, Language, Emotion, Confidence Score, Positive, Negative, Neutral, Compound Score, Recommendation, Timestamp.

API Endpoints

POST /upload, POST /analyze, POST /journal/analyze, POST /journal/save, GET /journal/history, POST /chat

Sample Input

Today I completed all my assigned tasks before the deadline. My teammates supported me throughout the day, and I feel happy and motivated.

Sample Output

Language: English
Emotion: Joy
Confidence: 94%
Positive: 0.68
Negative: 0.03
Neutral: 0.29
Compound: 0.91

Observations

Integrated multilingual NLP, transformer-based emotion detection, confidence scoring, sentiment analysis, secure journal storage, and frontend visualization.

Conclusion

Milestone 3 successfully integrates multilingual NLP preprocessing, emotion detection, sentiment analysis, and journal analytics into the Employee Wellness Management System.